NEXVIA AEGIS V9-M4 Methodological Audit

Final Common-Random-Numbers Semantics Determination and Hypothesis H3 Decision-Rule Aggregation Resolution

Autonomous Verification Subsystem (Google DeepMind Antigravity)

Git Branch: v9.0-runtime-assurance · *Implementation Commit: dc39f36* · *Date: October 25, 2026*

CRITICAL NON-AUTHORIZATION DIRECTIVE:
This audit document resolves the final methodological qualifications concerning the preregistered Common Random Numbers (CRN) semantics and Hypothesis H_3 decision rules. **This document does NOT constitute authorization to execute Milestone V9-M4.** All evaluations in this report were conducted strictly using synthetic test fixtures, in-memory models, and mock environments. Production holdout manifests (`test_mix_manifest_r1.json` and `test_scen_manifest_r1.json`) and output directories (`results/v9/test_mix/` and `results/v9/test_scen/`) do not exist. Reserved holdout seed blocks (73000..74439 for mixture holdout and 75000..75119 for scenario holdout) remain completely uninstantiated and untouched. Live execution requires explicit authorization from the project owner.

Abstract

This audit resolves the final two methodological questions concerning the NEXVIA AEGIS Milestone V9-M4 confirmatory holdout pipeline: (1) Does the implemented Common Random Numbers (CRN) architecture conform to the authoritative preregistered protocol, and what exact variance-reduction guarantees does it substantiate? (2) How are the primary V8.1 Martingale baseline and the co-evaluated V7 MPC baseline aggregated and interpreted under Hypothesis H_3? Through comprehensive citation of governing documents (`docs/v9_experiment_protocol.md`, `docs/v9_research_audit.md`, and `docs/v9_risk_calibration_design.md`), we establish that the authoritative protocol preregistered **Model A** (episode-level matched pairs with identical seed derivation, bitwise matching initial conditions at $t = 0$, and stateful streams coupled via common random numbers that diverge in closed loop as actions and queries differ). The alternative interpretation (**Model B**, requiring timestep- and channel-aligned exogenous noise independent of controller action) was never specified in the protocol; narrative statements in prior Report 14 suggesting timestep-wise identical disturbance across controllers were overstatements of report text. The implemented simulation architecture in `experiments/v9_evaluate_holdout.py` accurately fulfills Model A, providing valid paired-sample variance reduction for H_2 and H_3. Regarding Hypothesis H_3, we establish that Section 3.1.1 formally preregisters paired True-Success Rate (TSR) noninferiority against **V8.1 Martingale** as the primary confirmatory null hypothesis ($H_{0,3}$) at margin $\Delta_{\text{NI}} = -0.05$ on the S_{10} block, subject to nominal utility ($\widehat{\text{TSR}}_{\text{nominal}} \geq 0.94$). Comparison against **V7 MPC** is an auxiliary co-evaluated baseline. We refined `evaluate_scenario_holdout` to report `primary_passed`, `coeval_passed`, and `all_baselines_passed` without conflating their distinct statistical roles. Six additional synthetic unit tests (`test_50` through `test_55`) verify independent generator states, anchor-query offset mechanics, extension scenario exclusion, and all combinations of primary and co-evaluation pass/fail outcomes. All 55 preflight tests and all 166 full repository regression tests passed with zero failures.

Keywords: Common Random Numbers (CRN), Model A Coupling, Hypothesis H3, Dual-Baseline Aggregation, Decision Rule Resolution.

1 Executive Summary

1.1 Executive Summary (English)

1. **Registered CRN Semantics (Model A Confirmed):** The authoritative protocol defines paired comparisons at the episode level over matched (s, seed) pairs. As documented in `docs/v9_research_audit.md:240`, controllers initialize independent generators from identical seed derivations, matching initial target locations $\mathbf{p}_{\text{target}}$ and initial sensor noise at $t = 0$. In closed-loop execution, mutable streams advance statefully, and differing controller actions or anchor queries induce offsets in subsequent steps. Model B (spatio-temporal hash grids) was never preregistered.

2. **Clarification of Narrative Overstatements:** Report 14 contained narrative claims that CRN "guarantees identical physical disturbances and sensor noise across controllers at every timestep." We formally qualify this statement: identical noise holds strictly at initial conditions ($t = 0$) and across open-loop playback, but stateful streams offset when anchor queries differ. The paired statistical tests remain asymptotically and nonparametrically valid because episode pairs are i.i.d. and positively correlated.

3. **Exact H3 Baseline and Aggregation Rules:**
   - **Primary Baseline:** V8.1 Martingale (`AegisNextController`). Rejection of $H_{0,3}$ requires 1-sided 95% paired LCB of $\Delta\text{TSR}_{\text{V8.1}} > -0.05$ on the 10 scenarios of S_{10} **and** $\widehat{\text{TSR}}_{\text{nominal}} \geq 0.94$.
   - **Co-Evaluated Baseline:** V7 MPC (`AegisV7MPCBaselineController`). Evaluated at the same noninferiority margin $\Delta_{\text{NI}} = -0.05$.
   - **Aggregation Transparency:** The evaluator reports `primary_passed`, `coeval_passed`, `nominal_tsr_criterion` `all_baselines_passed`, and `passed` (tracking the primary null) without conflating co-evaluation into an ad-hoc joint intersection null.
4. **Synthetic Verification Expansion:** Implemented tests `test_50` to `test_55` in `tests/test_v9_holdout_preflight.` verifying: (i) independent generator memory state; (ii) exact offset realignment after anchor queries; (iii) primary pass with co-eval fail; (iv) primary fail with co-eval pass; (v) nominal TSR failure blocking both claims; and (vi) exclusion of extension scenarios S_{ext} from H_3.
5. **Regression Suite Integrity:** Focused preflight suite achieved 55/55 passed (1.11s); full repository regression suite achieved 166/166 passed (26.04s).
6. **Strict Holdout Non-Execution:** Zero real holdout manifests or output directories exist; reserved seed ranges (73000..74439 and 75000..75119) remain untouched; frozen policy SHA-256 (`5a89d606...`) is 100% unchanged.

1.2 (Executive Summary in Thai)

1. **การยืนยันแบบจำลอง CRN ทางการ (Model A):** พิธีสารหลักกำหนดการเปรียบเทียบแบบจับคู่ในระดับตอน (Episode-level matched pairs) บนคู่ (s, seed) โดยตัวควบคุมแต่ละตัวจะเริ่มต้นตัวสร้างเลขสุ่มอิสระจาก SeedSequence เดียวกัน ทำให้พิกัดเป้าหมายเริ่มต้ $(\mathbf{p}_{\text{target}})$ และสัญญาณรบกวน ณ ก้าว $t = 0$ เหมือนกันระดับบิต แต่ในระบบวงรอบปิด ลำดับสุ่มจะเคลื่อนไปตามการกระทำและการเรียกค้นเซนเซ Anchor ซึ่งตรงตามที่ระบุไว้ในเอกสารพิธีสารเดิมทุกประการ มิได้มีการกำหนดให้ต้องใช้แบบจำลองสุ่มแยกตามพิกัดเวลาและช่องสัญญาณ (Model B)
2. **การแก้ไขข้อความที่ระบุเกินจริง:** รายงานฉบับที่ 14 มีการระบุในเชิงบรรยายว่า CRN รับประกันสภาวะรบกวนที่เหมือนกันทุกก้าวเวลา ซึ่งข้อความนี้ได้รับการปรับปรุงและระบุข้อจำกัดอย่างชัดเจนว่า ความเหมือนกันระดับบิตเกิดขึ้น ณ สภาวะเริ่มต้น ($t = 0$) ส่วนการทดสอบทางสถิติ (H_2, H_3) ยังคงมีความถูกต้องทางสถิติและลดความแปรปรวนได้อย่างสมบูรณ์เนื่องจากคู่ตัวอย่างมีความสัมพันธ์เชิงบวกและเป็นอิสระต่อกันระหว
3. **ข้อกำหนดเกณฑ์การตัดสินของสมมติฐาน H3:**
   - **Baseline หลัก (Primary):** V8.1 Martingale โดยการปฏิเสธสมมติฐานว่าง $H_{0,3}$ ต้องได้ค่าขอบเขตล่างด้านเดียว 95% ($\text{LCB}_{95\%}$) ของ $\Delta\text{TSR}_{\text{V8.1}} > -0.05$ บนกลุ่ม S_{10} ร่วมกับเกณฑ์ความสำเร็จในสภาวะปกติ ≥ 0.94
   - **Baseline ประเมินร่วม (Co-evaluated):** V7 MPC ซึ่งประเมินที่ระยะความไม่ด้อยกว่าเดียวกัน ($\Delta_{\text{NI}} = -0.05$)
   - **ความโปร่งใสในการรายงานผล:** ปรับโครงสร้างรายงานให้แสดงผลลัพธ์แยกกันชัดเจน ได้แก่ `primary_passed`, `coeval_passed`, `all_baselines_passed` โดยไม่นำผลของ Baseline ร่วมไปปะปนกับสมมติฐานหลัก
4. **การขยายชุดทดสอบเชิงสังเคราะห์:** เพิ่มการทดสอบ `test_50` ถึง `test_55` ใน `tests/test_v9_holdout_preflight.py` เพื่อพิสูจน์ความเป็นอิสระของสถานะหน่วยความจำ, การเหลื่อมของสตรีมเมื่อเรียกค้น Anchor, และกรณีผลลัพธ์ผสมทุกรูปแบบ
5. **ความสมบูรณ์ของการทดสอบทั้งระบบ:** ชุดทดสอบ Preflight ผ่าน 55/55 รายการ (1.11 วินาที) และชุดทดสอบระบบรวมผ่าน 166/166 รายการ (26.04 วินาที)
6. **การคงความบริสุทธิ์ของข้อมูล Holdout:** ยืนยันว่าไม่มีการสร้างไฟล์ Manifest หรือรันซิมูเลชันบนข้อมูล Holdout จริง และค่าแฮชของนโยบายค 100%

2 Phase 1 — Registered CRN Semantics Determination

2.1 Authoritative Textual Evidence

The exact formulation of paired comparison in the governing documents defines pairing at the episode level:

- **Protocol Document (`docs/v9_experiment_protocol.md`):** Section 3.1.1 (line 30) preregisters:

  > *"One-sided 95% paired lower confidence bound (Newcombe/Wald paired difference CI on the 1,200 matched `(scenario, seed)` pairs) strictly exceeds* -0.05*..."*

  Section 3.2.4 (line 112) specifies the paired power analysis:

  > *"For paired binary `TSR` comparisons across N matched `(scenario, seed)` episodes with total discordant proportion* $q = p_{01} + p_{10}$*..."*

Table 3.5.1 (line 247) defines:

> *"Prespecified NI Test: 1-sided 95% paired LCB on* $\Delta TSR = TSR(\hat{\pi}) - TSR(\pi_{\text{ref}})$ *across* $N = 1,200$ *matched* $\mathcal{S}_{10}$ *pairs (*`D_test^scen`*)."*

- **Prior Research Audit (`docs/v9_research_audit.md`):** Line 240 provides the exact physical and mathematical definition of the coupling:

> *"every policy is evaluated on the exact same initial target position* `world.reset(np.random.default_rng(ep_seed)` *and identical sensor RNG seed* `np.random.default_rng(ep_seed)`*. Thus the 500 episodes form* ***500 matched pairs*** $(s, seed)$ *across any two policies* (π_A, π_B) ***(note that closed-loop trajectories diverge once actions differ, so sensor draws remain matched at step*** $t = 0$ ***and coupled via common random numbers thereafter)****. Statistical tests on episode-level binary or continuous outcomes across 500 episodes* ***must respect this paired design****."*

2.2 Model A vs Model B Formal Resolution

- **Model A (Preregistered Design):** Independent Generator instances are derived from matching `SeedSequence` keys. Initial state and initial measurements at $t = 0$ match bitwise. In closed loop, random numbers are drawn sequentially by mutable generators; differing actions and anchor query counts cause stream offsets in later steps.
- **Model B (Hypothetical Strict Grid):** Exogenous disturbances and sensor noise are indexed across a 4D spatio-temporal tensor $\mathbf{W}(s, \text{seed}, t, c)$ independently of controller behavior.

Authoritative Finding: The registered protocol formally specifies **Model A**. Model B was never preregistered or implemented. The narrative claims in Report 14 suggesting timestep-wise identical disturbance across controllers were overstatements of report text, now formally qualified.

3 Phase 2 — Verification of Randomization Guarantees

Table 1 decomposes the exact randomization properties supported by the codebase.

Table 1: Decomposition of Supported and Qualified Randomization Guarantees

ID	Guarantee Description	Supported?	Evidentiary Proof & Verification Basis
G1	Separate in-memory RNG state	**YES**	Generators derived independently; verified in `test_48` & `test_50`.
G2	Deterministic reproducibility	**YES**	Invariant to controller execution order; verified in `test_47`.
G3	Common initial conditions	**YES**	$\mathbf{p}_{\text{target}}$ and $t = 0$ sensor noise bitwise identical; verified in `test_49` & `test_50`.
G4	Timestep-aligned noise ($t > 0$)	**QUALIFIED**	Holds for open-loop; offsets occur when anchor queries differ (`test_51`).
G5	Paired variance reduction	**YES**	Positive correlation across matched episodes reduces $\text{SE}(\overline{\Delta\text{TSR}})$.

4 Phase 3 — Hypothesis H3 Decision-Rule and Aggregation Semantics

4.1 Formal Protocol Requirements

Under Section 3.1.1 of the Authoritative Protocol:

$$H_{0,3} : \text{TSR}(\hat{\pi}; \mathcal{P}_{10}) - \text{TSR}(\pi_{\text{V8}}; \mathcal{P}_{10}) \leq \Delta_{\text{NI}} = -0.05 \quad \text{(co-evaluated vs } \pi_{\text{V7}} \text{ at } -0.05). \tag{1}$$

The decision rule requires:

$$\text{LCB}_{95\%}(\Delta\text{TSR}_{\text{V8.1}}) > -0.05 \quad \text{AND} \quad \widehat{\text{TSR}}_{\text{nominal}} \geq 0.94. \tag{2}$$

Evaluation is restricted strictly to the 10 scenarios in $\mathcal{S}_{10}$ ($N = 1,200$). Extension scenarios $\mathcal{S}_{\text{ext}}$ (scenarios 11 & 12) are excluded from H_3.

4.2 Aggregation Architecture in the Evaluator

The evaluation dictionary in `experiments/v9_evaluate_holdout.py` provides complete transparency:

- `h3_overall["passed"]`: Tracks rejection of primary null $H_{0,3}$ against V8.1.
- `h3_overall["primary_passed"]`: Boolean tracking primary V8.1 noninferiority + nominal TSR.
- `h3_overall["coeval_passed"]`: Boolean tracking co-evaluated V7 noninferiority + nominal TSR.
- `h3_overall["all_baselines_passed"]`: Boolean tracking whether both baselines pass.
- `h3_overall["status"]`: Categorical status string indicating the exact joint configuration.

5 Synthetic Test Suite and Empirical Verification

Table 2 details the expanded test suite in `tests/test_v9_holdout_preflight.py`.

Table 2: Methodological and Synthetic Invariant Verification Tests

Test ID	Target Mechanism	Verified Invariant & Methodological Property	Time	Status
`test_47`	Order Invariance	V9/V8/V7 forward vs reverse order bitwise identity	0.06s	**PASS**
`test_48`	Termination Isolation	Early termination at $t = 5$ leaves other controllers unaffected	0.05s	**PASS**
`test_49`	Initial States & Offset	Bitwise $t = 0$ match; characterizes mutable stream offset	0.05s	**PASS**
`test_50`	Generator Independence	Distinct in-memory state; channel A noise match at $t = 0$	0.04s	**PASS**
`test_51`	Query Stream Shift	Proves exact scalar realignment after anchor queries ($k = 2$)	0.03s	**PASS**
`test_52`	Primary Pass / Co-eval Fail	`primary=True`, `coeval=False`, `passed=True`	0.06s	**PASS**
`test_53`	Primary Fail / Co-eval Pass	`primary=False`, `coeval=True`, `passed=False`	0.06s	**PASS**
`test_54`	Nominal TSR Barrier	Nominal TSR < 0.94 blocks both V8 and V7 claims	0.05s	**PASS**
`test_55`	S10 Domain Restriction	Altering S_{ext} outcomes leaves H_3 statistics invariant	0.06s	**PASS**

6 Regression Testing and Holdout Isolation Proof

6.1 Test Suite Results

- **Focused Preflight Suite:** `PYTHONPATH=src uv run pytest -v tests/test_v9_holdout_preflight.py` passed all 55 tests in 1.11 seconds.
- **Full Repository Suite:** `PYTHONPATH=src uv run pytest -q` passed all 166 tests in 26.04 seconds with zero regressions.

6.2 Proof of Production Holdout Isolation

A comprehensive filesystem scan verified that zero real holdout manifests or directories exist:

$$\begin{aligned} \text{Path Existence}(\texttt{results/v9/manifests/test_mix_manifest_r1.json}) &= \textbf{FALSE} \\ \text{Path Existence}(\texttt{results/v9/manifests/test_scen_manifest_r1.json}) &= \textbf{FALSE} \\ \text{Path Existence}(\texttt{results/v9/test_mix/}) &= \textbf{FALSE} \\ \text{Path Existence}(\texttt{results/v9/test_scen/}) &= \textbf{FALSE} \end{aligned} \tag{3}$$

The frozen policy SHA-256 remains `5a89d6061b379f1b34ab47be0deef8d63180d3e4c21d18c478bc7e451f5c84e2`.

7 Conclusion

The final status of this audit is formally declared as:

CRN_H3_PROTOCOL_CONFIRMED_WITH_QUALIFICATIONS

The CRN architecture conforms to the preregistered episode-level matched design (Model A), the H3 decision rules are transparently aggregated with primary and co-evaluated baselines distinctly reported, and the evaluation harness remains safely locked pending project-owner authorization.